

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (CSE) (Credit System) Degree Examinations**

January 2023

19CSE55 – BLOCKCHAIN TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) Discuss any four primary features of Blockchain with example.	10	L*3	1	1
	b) Define blockchain and elaborate its importance.	10	L4	1	1
2.	a) Discuss the applications of blockchain with respect to Healthcare System and Product Tracking.	10	L4	1	2
	b) Illustrate how Bitcoin can be used to solve the issues related to centralized banks.	10	L3	2	2
3.	a) Define Consensus Mechanisms and explain any three of them.	10	L3	2	2
	b) Summarize the benefits and limitation of blockchain.	10	L4	2	2
		Unit – II			
4.	a) Define Ethereum network and Elaborate Mainnet, Testnet and Private net.	10	L4	3	2
	b) Explain Ethereum ecosystem, along with components of Ethereum ecosystem.	10	L3	3	2
5.	a) Discuss Disintermediation and Competition methods to achieve decentralization.	10	L4	3	2
	b) Explain generic model of an Oracle and smart contract ecosystem with a neat diagram.	10	L3	4	2
6.	a) Illustrate the working of Ricardian contracts along with its properties.	10	L5	4	2
	b) Explain smart contract and its properties.	10	L3	4	2
		Unit – III			
7.	a) Discuss the Transaction flow architecture of Hyperledger fabric.	10	L4	5	2
	b) Elaborate Hyperledger fabric network along with its components.	10	L3	5	1
8.	a) Explain Chaincode Services with a neat diagram.	10	L3	5	2
	b) Define Hyperledger and Discuss its significant goals.	10	L4	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
